Historical Mathematical Models

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Project Background



One of the largest collections worldwide

Over 400 models made of wood, plaster, strings, etc.

Mostly made in Germany, with some produced in the United States.

Used to be located in Altgeld.

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Some examples include:

- Sphere
- Two types of Meissner tetrahedrons
- Surface of rotation created by spinning a reuleaux triangle around its axis of symmetry
- and more!

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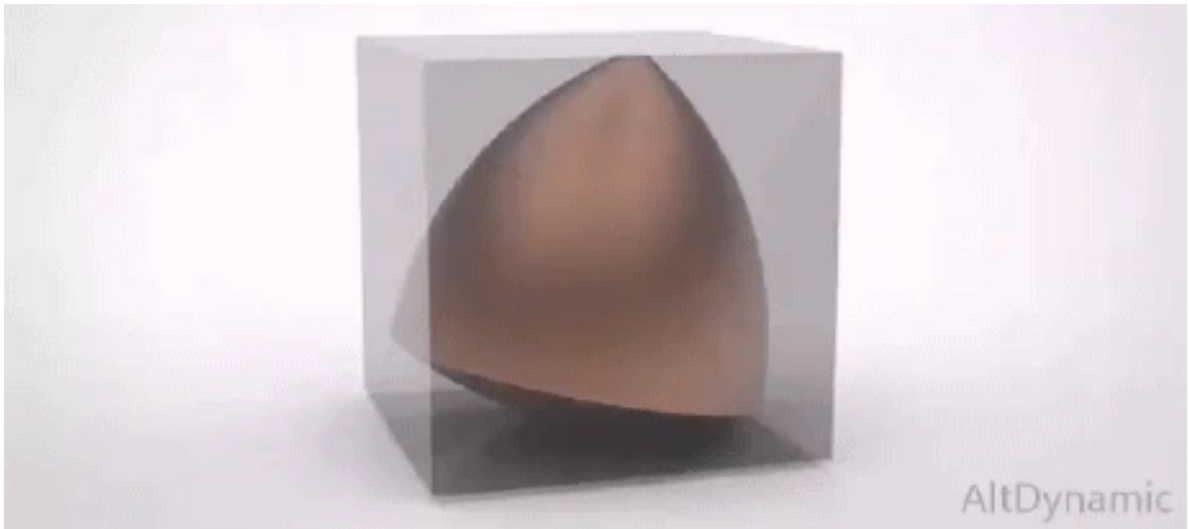
Hence, the "Digital" part of "Digital Library".

Where have we started?

Written Descriptions

Where have we started?





external gif link

Where have we started?

Schilling Series 40 No. 3: Meissner Tetrahedron

Written by Dashiell

A shape of constant width, which is a kind of Meissner Tetrahedron, a modification of a Reuleaux Tetrahedron. It is constructed as follows:

- Begin with a regular tetrahedron of side lengths.
- At some vertex, construct a section of a sphere of radius s which covers the opposite face
- Repeat 2 for each vertex.
- Smooth out three of the edges which meet at a single vertex by replacing them with a surface of rotation of a circular arc.

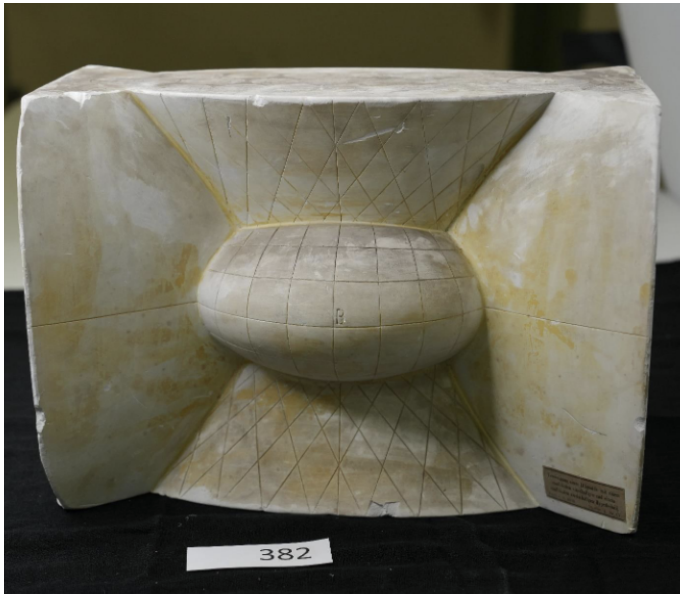
Written Descriptions

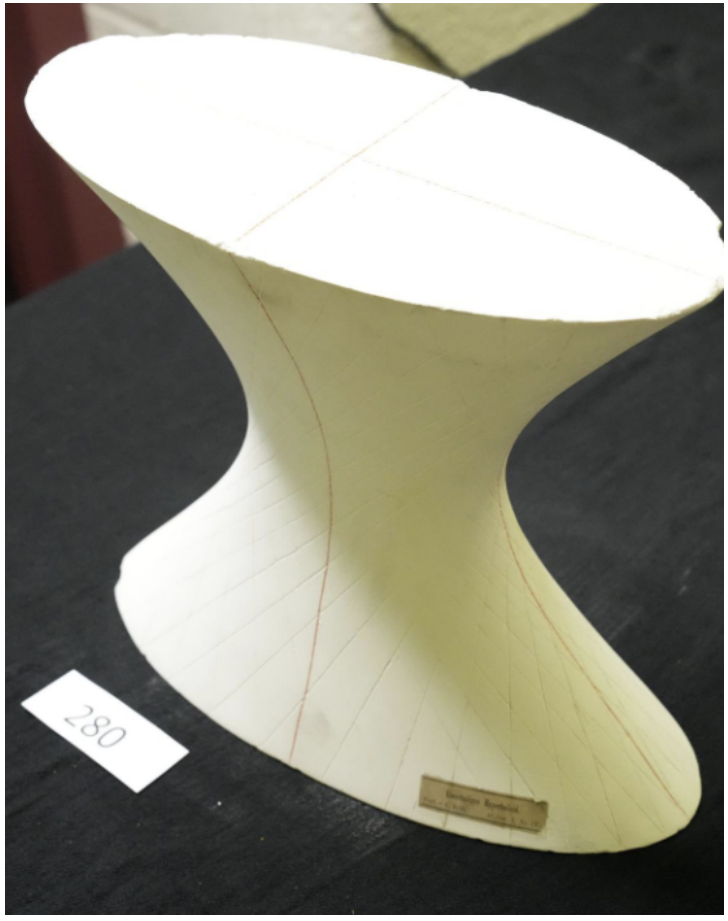


Schilling Series 16

Schilling Series 16





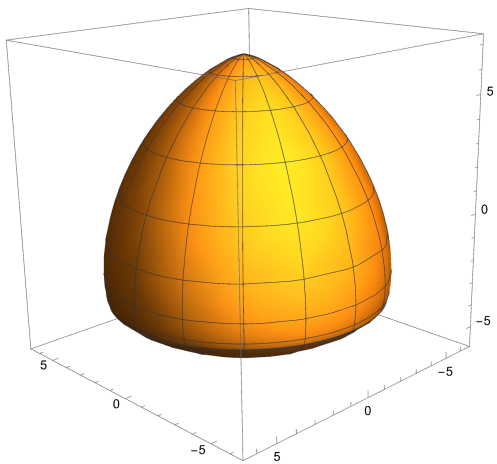


Modeling and 3D Printing

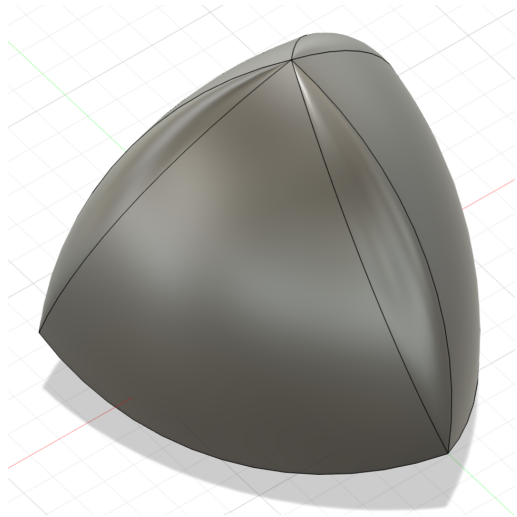
Digital models are generally constructed one of two ways:

- Wolfram Mathematica, for ones that are defined algebraically
- Autodesk Fusion 360, for ones that are defined geometrically

From there, they are exported to .STL files to be 3D printed.



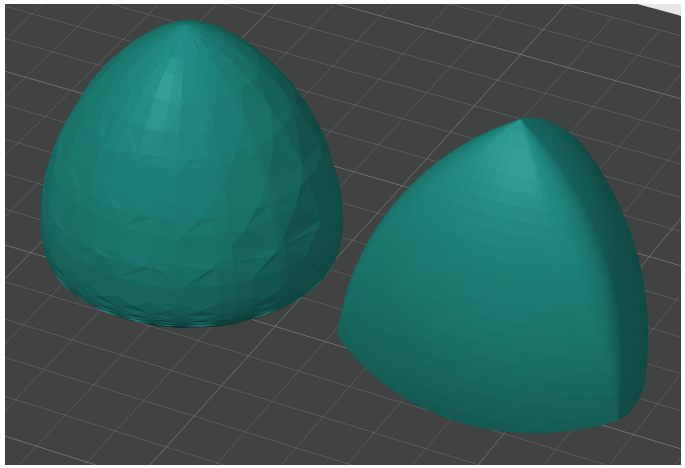
(a) Sch. 40, Model 1, defined with a paremetric equation, Mathematica



(b) Sch. 40, Model 3, defined as a Meissner tetrahedron, Autodesk

Figure: Two models from Sch. Series 40, modeled in different programs

Modeling and 3D Printing (cont.)



(a) Sch. 40, Models 1 and 3, in a slicing software to be printed

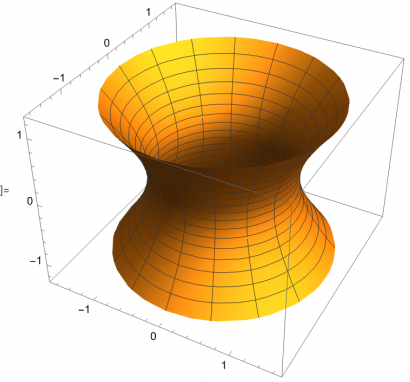
```
In[137]:= x[u_, v_] := a * Cosh[u] * Cos[v]
          y[u_, v_] := b * Cosh[u] * Sin[v]
          z[u_, v_] := c * Sinh[u]
          a = 1
          b = 1
          c = 1
          ParametricPlot3D[{x[u, v], y[u, v], z[u, v]}, {u, -1, 1}, {v, 0, 2 Pi}]
```

```
Out[140]= 1

Out[141]= 1

Out[142]= 1
```

binary form | prime? | perfect number? | Roman numerals | more...



(b) Sch. 16, Nr 4

Figure: STL Files from Fusion are far smoother, but Mathematica allows for more options in modeling.

Outreach

Practical Extension of the Project: Outreach
Create a hands-on approach to geometry with activities such as:

- Physically playing around with the models
- Show the more artistic properties of Reuleaux polygons
- Visual demo of curves of constant width



Figure: Einschaliges Hyperbolid from Schilling Series 16. A basic surface

Conclusion

Began describing models from Series 40 and 16.

Started making .STL files to 3d print.

Began writing outreach plans to educate students in the community.

Overall, we plan to not only catalog the historical models, but use them for their original purpose: to educate students on 3d surfaces.